WEEK 4 · LECTURE L3 OF 3 · UNIT II

Round 2: The Architecture

Designing the stack: the module landscape, sequencing, and the anchor test.

Part 1 GreenLeaf's 16 systems in three layers

Part 2 Sequencing under budget and capacity limits

Part 3 The anchor test, and a look back at Round 2

Read Your pod's Round 2 submission and D1

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L3 · STACK DESIGN

GreenLeaf's 16 systems, sorted into three layers

Layer	Systems (adoption rate)	What it tells you
Record	LegacyHR 0.43 · PeopleCloud pilot 0.38 · PayEasy Payroll 1.00 · DocVault 0.47	Two core HRIS compete; only payroll is fully used
Engagement	HireRight ATS 0.79 · ESS App 0.41 · ShiftPlan 0.40 · GreenLearn LMS 0.37 · SahayakBot 0.33 · ReferWin 0.33 · PulseEngage 0.30 · SkillGraph LXP 0.30 · WellbeingPlus 0.18 · ExpenseFlow 0.50	Many tools, mostly under a half of licensed users
Insight	InsightHR Analytics 0.42 · Spreadsheet Trackers 1.00	Analysis still runs on spreadsheets

SOURCE TechStack_Inventory, columns B, C and I. The Record, Engagement, Insight framing is the course's three-layer architecture framing.

PART 1 · LICENSE COST AGAINST USE

What GreenLeaf pays for each active user

System	Annual license	Active users	Cost per active user
InsightHR Analytics	₹17,00,000	38 of 90	₹44,737
PeopleCloud HRIS (pilot)	₹42,00,000	150 of 400	₹28,000
HireRight ATS	₹15,00,000	95 of 120	₹15,789
SkillGraph LXP	₹14,00,000	360 of 1,200	₹3,889
GreenLeaf ESS App	₹11,00,000	2,900 of 7,000	₹379
PayEasy Payroll	₹13,00,000	8,000 of 8,000	₹162

READ IT

Cost per active user = Annual_License_Cost_INR ÷ Active_Users. Low adoption makes even a cheap license expensive. Source: TechStack_Inventory.

L3 · SEQUENCING

Order the stack under budget and capacity limits

1 · Dependency

Record before engagement before insight. An app or dashboard built on two competing records reports two truths.

2 · Absorption

People can absorb only so much change at once. Each go-live spends Change Capital and trainer time.

3 · Obligation

Consent, data residency and retention under the DPDP Act cannot wait for a later phase.

The budget

₹6 crore covers licenses, implementation and change management for all six rounds. Current licenses alone cost ₹1.845 crore a year.

The reach test

Which anchor does each phase reach first? A plan that serves only laptop users in Year 1 defers the majority.

ASK OF ANY PLAN What has to be true first, who carries the change, and what are we stopping to pay for it?

PART 3 · THE ANCHOR TEST

Does your stack work for all four people?

GL-1001 Lakshmi

Can she use it on her phone, in Telugu, with weak data? If not, it is not live for her.

GL-1002 Arjun

Gig partners are outside Employee_Master. Which system, if any, sees him at all?

GL-1003 Sunil

Which tasks does the stack take from him, and what new role does it offer him?

GL-1004 Shreya

Does she get one clean record and usable analytics, or another spreadsheet?

THE RULE A stack that passes for Shreya and fails for Lakshmi serves a minority of GreenLeaf's workforce.

PART C · LOOK BACK AT ROUND 2

All four pods chose Option 2A

The record

Which system did your stack name as the system of record?

The duplicate

Did your plan retire LegacyHR, PeopleCloud, or neither?

The ATS

Where did the ATS sit in your stack, and who checked it before Round 4?

Going deeper: the theory underneath architecture

The next slides set out Ross, Weill and Robertson, modularity, Conway's law, and path dependence: what each claims, how it works, where it breaks, and what it means at GreenLeaf.

THEORY IN DEPTH · ROSS, WEILL AND ROBERTSON (2006)

Architecture follows the operating model

THE CORE IDEA

Firms differ on how much they standardize processes and how much they integrate data.

HOW IT WORKS

Four models: diversification, coordination, replication, unification. Each implies a different architecture.

LIMITS AND CRITIQUE

Real firms mix models across units, so the choice is rarely pure.

AT GREENLEAF 180 similar stores suggest replication: standardize the store process and copy it everywhere.

THEORY IN DEPTH · BALDWIN AND CLARK (2000)

Modularity creates options

THE CORE IDEA

A modular system splits into parts that connect through stable interfaces.

HOW IT WORKS

Modules can be swapped or improved independently, which creates option value.

LIMITS AND CRITIQUE

Interfaces must be designed well up front; poor interfaces make integration costly.

AT GREENLEAF

A modular core with clean interfaces lets GreenLeaf replace the ATS later without rebuilding the record.

THEORY IN DEPTH · CONWAY (1968)

Systems mirror the organization that builds them

THE CORE IDEA

Conway's law: a system's design copies the communication structure of the organization.

HOW IT WORKS

Separate teams produce separate systems that do not talk to each other.

LIMITS AND CRITIQUE

An observation rather than a law; teams can deliberately design across their boundaries.

AT GREENLEAF Sixteen disconnected systems mirror an HR function grown by acquisition, team by team.

THEORY IN DEPTH · ARTHUR (1989)

Early choices lock in later ones

THE CORE IDEA

Path dependence: small early choices become self-reinforcing through learning, networks and switching costs.

HOW IT WORKS

Once data and habits build around a system, leaving it costs more each year.

LIMITS AND CRITIQUE

Lock-in can be broken by a large enough shock or a much better alternative.

AT GREENLEAF

LegacyHR's 2010 spreadsheets persist because every workaround depends on them. Plan the exit, not just the entry.

THREE TAKEAWAYS

1. Name the system of record first; every module choice follows from it.

2. Integration is the cost that never shows on the license invoice.

3. Audit who an ATS screens out, at every stage, before trusting its speed.

READ BEFORE NEXT CLASS

Readings for this session

Bondarouk, Parry and Furtmueller (2017)
Electronic HRM: Four Decades of Research. IJHRM, 28(1), 98 to 131.

Caldwell (2022)
HR Analytics and Digital HRM. Chapter on HRIS and digital HR systems.

Kavanagh and Johnson (2017)
Human Resource Information Systems, 4th ed. System of Record, Engagement and Insight.